Technical Documentation Version 9.6

Building a Model

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Chapter 1
Model Building

Topics

Model Building Process Overview

Building a RiverWare model typically consists of the following general steps

1. Select the Simulation Controller and set run times for a simulation run.
2. Create objects on the workspace.
3. Configure objects with name and method selections.
4. Define your desired unit scheme.
5. Set timeseries ranges. Enter, import, and export slot data.
6. Link objects to define the basin topology.

This document describes these steps and provides links to documents with more information.

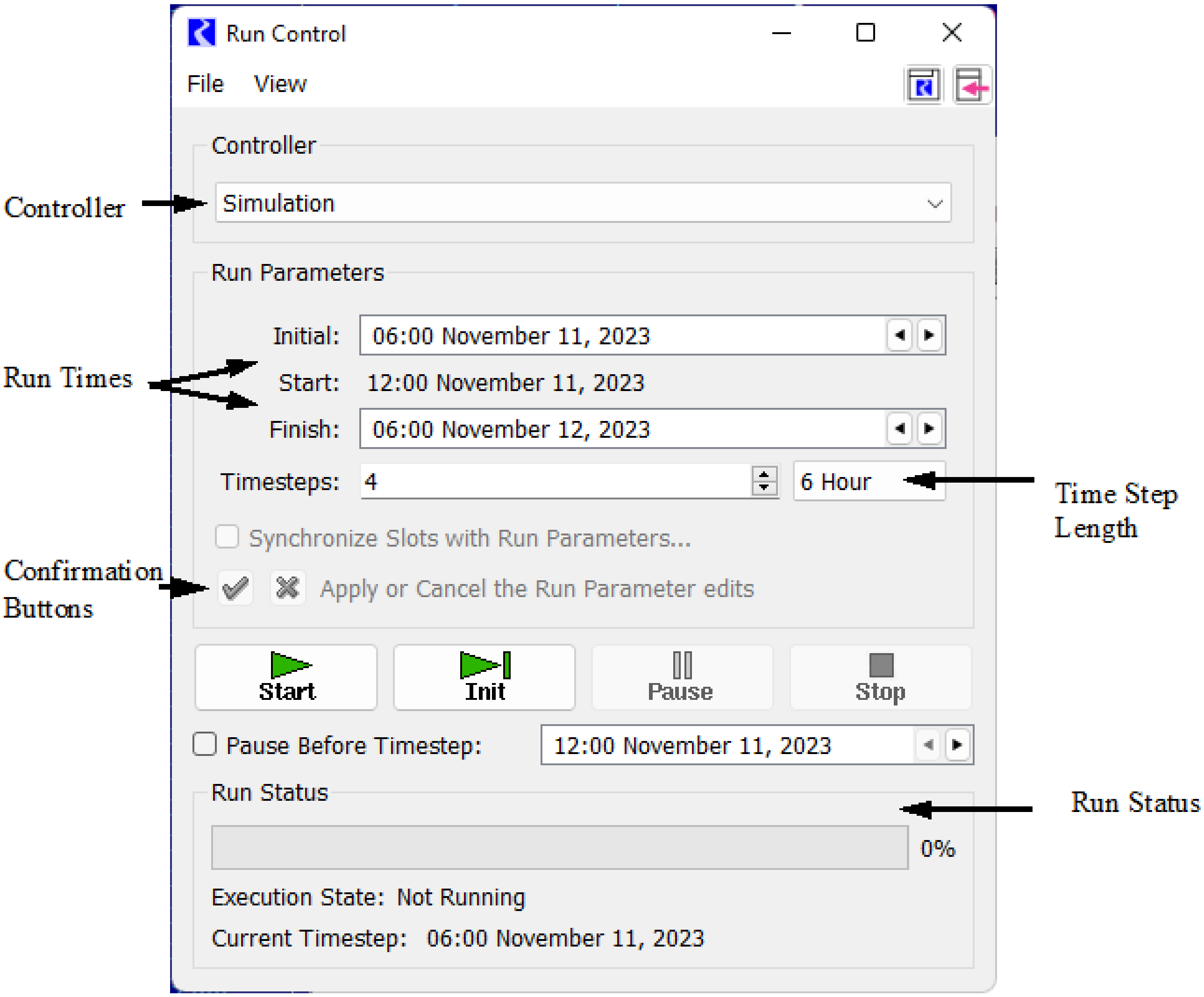
Selecting the Controller and Setting Run Times

The first step in building a model is to select the controller and specify the run times and timestep for the run. These actions are performed from the Run Control. You can use either of the following methods to open the Run Control.

- On the workspace menu, select **Control**, then **Run Control Panel**.

- On the workspace toolbar, select **Run Control**. 

Figure 1.1 Run Control Annotated Screenshot



On the Run Control, specify the following settings:

- Controller
- Timestep length
- Initial date
- Finish date or number of timesteps

See “Controllers” in *User Interface* for a detailed description of this window.

Creating Objects on the Workspace

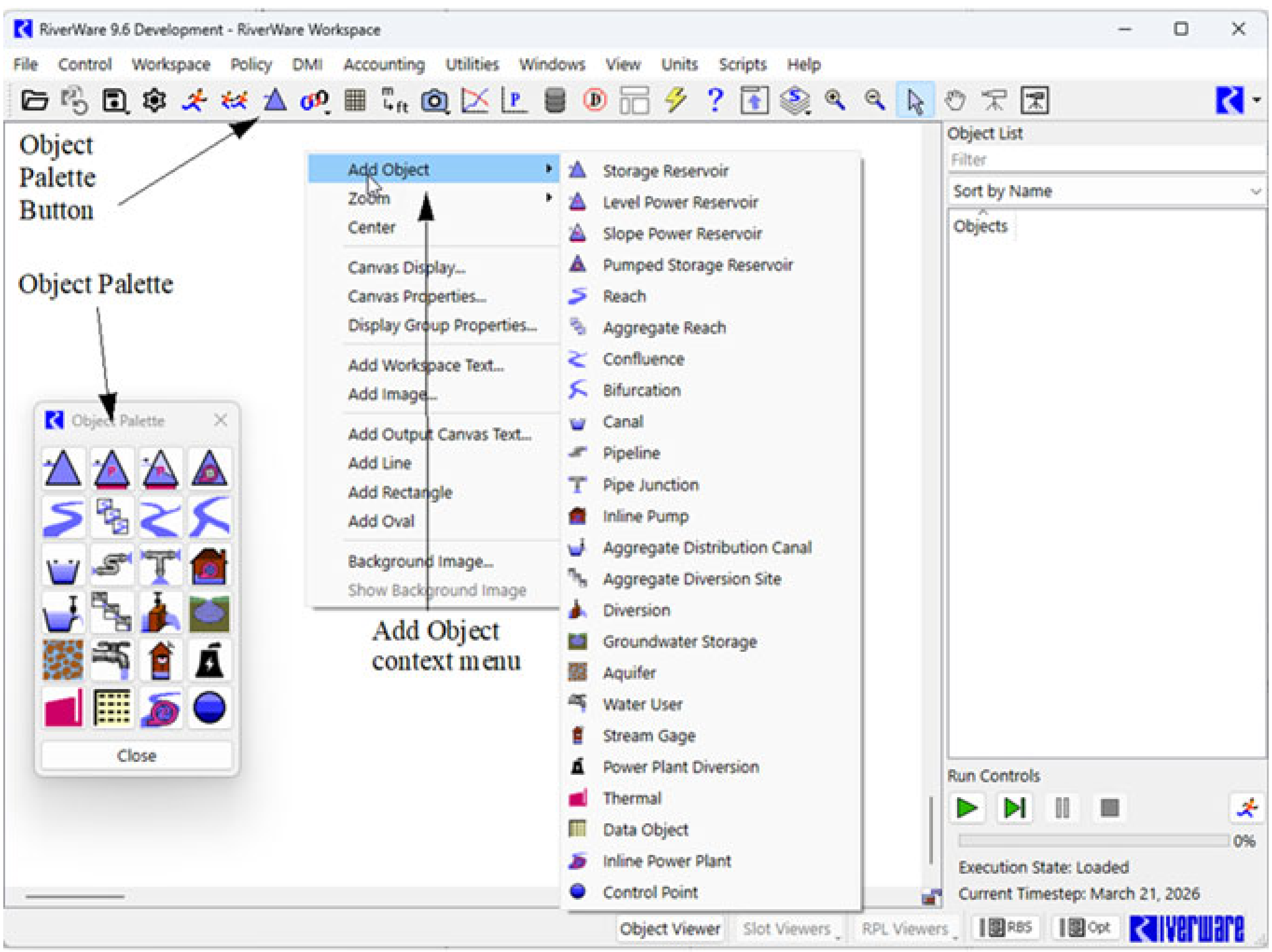
You can create a new model by placing object icons on the workspace. By positioning the icons to reflect the objects' relative physical locations, you can create a workspace model that helps you to visualize the physical layout of the modeled system. Objects also include the data and the physical process algorithms that drive the simulation.

> **Note:** The coordinates of the objects on the workspace are saved with the model so objects will retain their designated positions when the model is reopened.

You can use either of the following methods to create objects on the workspace; Figure 1.2 illustrates.

- Pull objects from the Object Palette; see "Creating Objects Using the Object Palette" on page 4.
- Use the Add Object option on the workspace shortcut menu; see "Creating Objects Using the Workspace Shortcut Menu" on page 4.

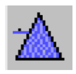
Figure 1.2 Approaches to adding objects onto the RiverWare workspace



Creating Objects Using the Object Palette

The Object Palette includes all available object types used to represent features of a river basin. To place objects on the workspace, drag them from the Object Palette to the desired location on the workspace.

You can use either of the following methods to open the Object Palette.

- On the workspace toolbar, select **Object Palette**. 
- On the workspace menu, select **Workspace**, then **Objects,** then **Object Palette**.

Creating Objects Using the Workspace Shortcut Menu

The Add Object option on the workspace shortcut menu includes the same object types as the Object Palette.

To add objects using this method, right-click in the workspace where you want to add an object, then in the shortcut menu, select **Add Object**, then select the object.

Default Object Attributes

At the time an object is created, default attributes are defined by several sources, as follows.

- The first timestep and timestep size in series slots are matched to the run control run time settings.
- Display units and other slot configurations conform to the currently active unit scheme. See “Unit Schemes” in *User Interface* for details.
- Default user methods are selected for new objects. Slots associated with the selected controller and methods are allocated in memory. See “Object Types” in *Objects and Methods* for details on each type of object.

See “Objects on the Workspace” in *User Interface* for details on managing objects, including selecting, importing, exporting, deleting, and clearing the workspace.

Configuring Objects

Objects are configured from the Object Viewer or Open Object window. See “Object Viewer and Open Object” in *User Interface* for details.

The general steps for configuring objects are as follows.

1. Name the object. See “Naming an Object” in *User Interface*.
2. Select user methods. See “Selecting Methods” in *User Interface*.

   Note: The Multi-Object Method Selector can be used to select methods on multiple objects in one action. See “Multiple Object Method Selector” in *User Interface* for details.

Defining Units

RiverWare stores all slot values in RiverWare internal units and performs calculations in these units, except where the algorithms do explicit conversions. You can display values in any user unit you want. You define the user units, scale, precision, and format by setting up a *unit scheme*. See “Unit Schemes” in *User Interface* for details on features of unit schemes.

Importing and Entering Data Into Slots

The next step in model building is to enter data into slots. See “Slot Windows” in *User Interface* for details on the slot interface. In general, enter necessary table, scalar and periodic data first. Then add series data that will drive the simulation. Data can be entered in the following ways

- Keyboard entry
- Copy and paste from text files or spreadsheets. In RiverWare, this is typically considered an “Import Paste” when the data is coming from the system clipboard. See “Import Paste” in *User Interface* for more information.
- Import from a text file
- Import from a database or spreadsheet using the Data Management Interface. See “Data Management Interface (DMI)” for more information.

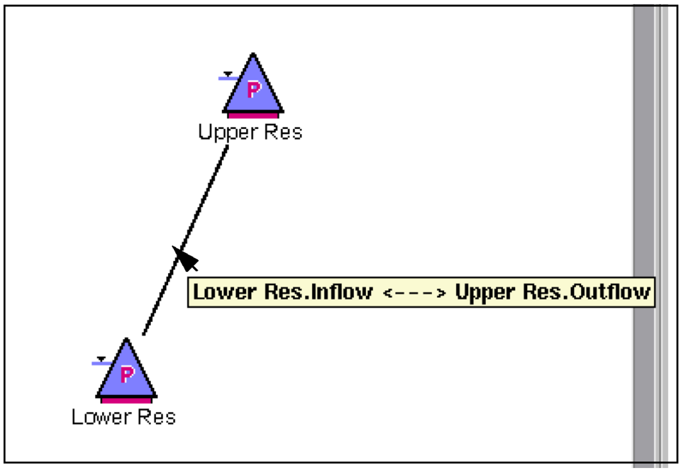
Linking Slots on Objects

Links are the connections between slots on objects. Links pass information during a simulation run; they propagate a value from a slot on one object to a slot on a different object. Links are generally bidirectional, but in a pure simulation run, if values propagate in both directions during a single timestep, the model is overdetermined and the run quits prematurely.

A slot may be linked to more than one other slot. Also, there may be multiple links between different slots on two objects, but only one line is displayed on the workspace.

Viewing Existing Links on the Workspace

In models with many links, it can be difficult to determine which objects and slots a particular link is connecting. When you hover over a link, all the connected objects and slots associated with the link are displayed in a tooltip, as shown in the following figure. The link name is also shown in the workspace status bar (lower-left corner). The status bar display is useful because the link name is displayed there for as long as you hover over the link, whereas the tooltip shows the name for only a few seconds.



On the Object Viewer (or Open Object) **Slots** tab, linked slots are marked with the Linked symbol. To display a list of all slots linked to a particular slot, right-click the linked slot, then select **Linked Slots** in the shortcut menu. Select a slot on the shortcut menu to open that slot.

Editing Links

The following utilities allow you to view, create, and delete links.

- Shortcut menus—create links between any two slots on the workspace; delete any existing links from the workspace.
- Smart Linker—create recommended links between two objects; delete existing multiple links between two objects in one operation.
- Edit Links tool—create a link between any two slots; delete any existing link.

These utilities are described in the following sections.

> **Tip:** You can also create and delete links in a RiverWare script. See “Link Slots” in *Automation Tools* for details.

Shortcut Menus

You can use right-click shortcut menus on the workspace to create or delete links.

Creating Links

The Link Slots option on the shortcut menu allows you to link any two slots. To create a link, right-click an object; then in the shortcut menu, select **Link Slots**, then the slot to link. A link is started. Next, right-click the object where the link will end, select **Link Slots** on the shortcut menu, then select the slot on that object. The link is created.

> **Note:** Using this method, you can link any two dispatch slots in the model regardless of whether the link makes physical sense; this practice is not recommended.

Deleting Links

To delete a link between objects, hover over the link and right-click. On the shortcut menu, select **Delete Link**, then select a link from the list. The link is deleted.

Smart Linker

The Smart Linker simplifies the process of linking slots by recommending a set of links between two selected objects based on the type of object, their method selections, and the relationship between the two objects.

> **Note:** The recommended links have been defined by CADSWES based on common modeling practices, experience, and existing models. You may want to link other slots, however, in which case you would not use the Smart Linker.

For example, if you select a Reach and a Groundwater Storage object, and the Head Based Seepage method is selected on the Reach and the Head Based Groundwater Grid method is selected on the Groundwater Storage object, the Smart Linker proposes the following links:

- Reach.Previous Water Table Elevation to Groundwater.Elevation Previous and Reach
- Seepage to Groundwater.Inflow from Surface Area

If you are also modeling evaporation or salinity, the Smart Linker proposes another set of links.

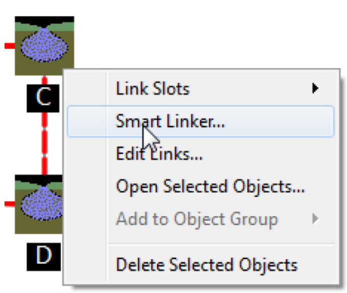
Opening the Smart Linker

To open the Smart Linker, select two objects on the workspace, then do one of the following.

> **Note:** You must select exactly two objects; if you select more than two, a warning dialog opens.

- Right-click and select **Smart Linker**.
- On the workspace toolbar, select **Link** , then **Smart Linker**.
- On the workspace menu, select **Workspace,** then **Smart Linker**.

If the two selected objects do not have any defined recommended links, a warning dialog opens. If you want to link slots on these objects, use the Link Slots shortcut menu or the Edit Links tool.

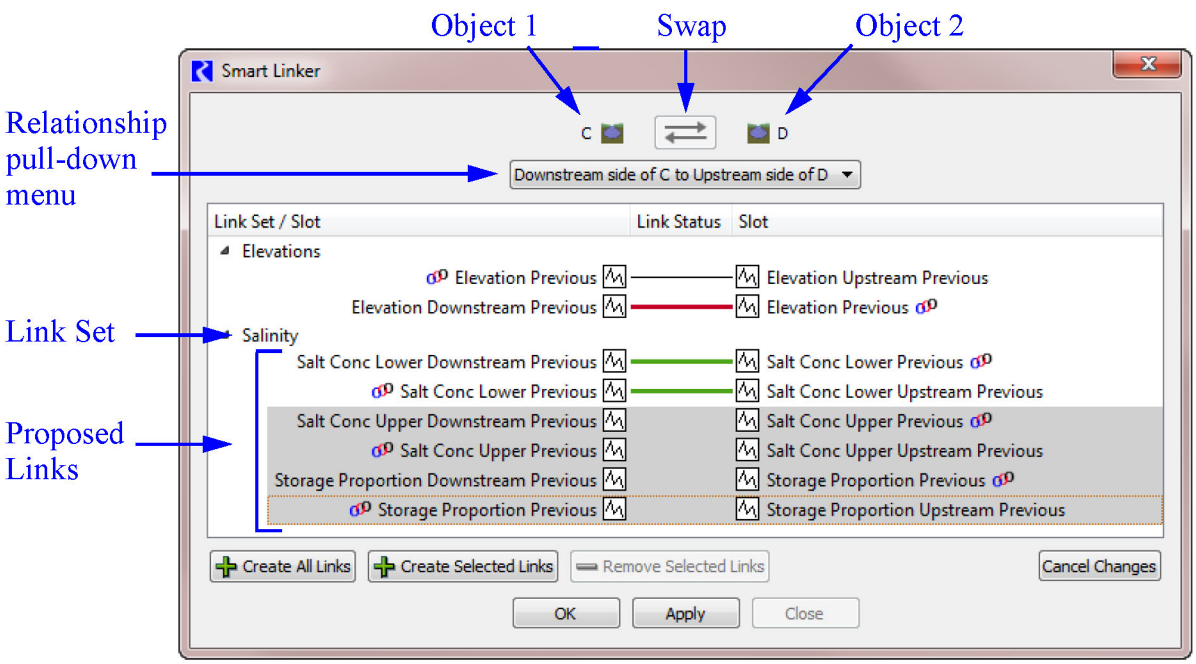


About the Smart Linker

Figure 1.3 shows the Smart Linker, which is described below.

> **Note:** Also, see the Smart Linker video for a demonstration of its use: https://riverware.org/tutorials/SmartLinker/index.html

Figure 1.3 Main regions of the Smart Linker



The Smart Linker displays the two selected objects at the top. The Swap button  swaps the two objects, left to right. It is disabled if there are unapplied link additions or removals, or if the reversed order is not supported.

The relationship menu lists the available spatial relationships between the two selected objects. Use this menu to indicate the relative position or relationship of the two objects. For example, is Object 1 upstream or downstream of Object 2? Possible options include: upstream / downstream; left / right; diverts from / diverts to; returns from / returns to. The relationship you select leads to different recommended links.

In the body of the Smart Linker, sets of recommended links are listed in a tree view with the specific slots shown.

Table 1.1 lists the meaning of each type of horizontal line that may appear in the Link Status column.

Table 1.1 Link line legend

Image	Color	Description
	thin black	Existing link
	thick green	Tentatively added link
	thick red	tentatively removed link

Buttons along the bottom allow you to create or delete selected links. These changes are made tentatively. You can select **OK** or **Apply** to complete the operation or **Cancel Changes** to cancel the changes.

Selecting Aggregate/Element Objects

Aggregate objects are handled a little differently from other objects. Because the Smart Linker is opened by selecting two objects on the workspace, when you select an aggregate object, there is no way for you to select an element; you can, however, select the element in the Smart Linker.

When an aggregate object is shown in the Smart Linker, the Element checkbox is displayed. To link the aggregate object, leave the **Element** checkbox cleared. To link the element object, select the **Element** checkbox, then select the element. For example, in the following figure, the Reach0 reach element will be linked to the aggregate diversion site.

Creating Links

To link the slots shown in a particular row, highlight the row, then select **Create Selected Links**. Alternatively, select **Create All Links** to create all recommended links. The Link Status for the row then turns to a green line. Select **OK** or **Apply** to confirm creation of the links.

Deleting Links

You can delete any existing link by highlighting the row and selecting **Remove Selected Links**. The Link Status for that row turns to a red line. Select **OK** or **Apply** to confirm deletion of the link.

Using the Smart Linker

Following are the basic steps for using the Smart Linker to create all recommended links between two objects.

1. Select two objects on the workspace that you want to link. Right-click and select **Smart Linker**.
2. In the Smart Linker, use the **Swap** button to orient the two objects as you want.
3. If applicable, use the relationship menu to specify the relation ship between the two objects. After each change, a different set of slots is listed as recommended links.
4. Select **Create All Links** to create all recommended links.
5. Select **OK** to confirm creation of the links and close.

Edit Links Tool

In some situations, the link operations described in the previous sections may not provide enough flexibility for you to view, create, or edit links. In these cases, the Edit Links tool provides a central location for you to perform these activities.

Opening the Edit Links Tool

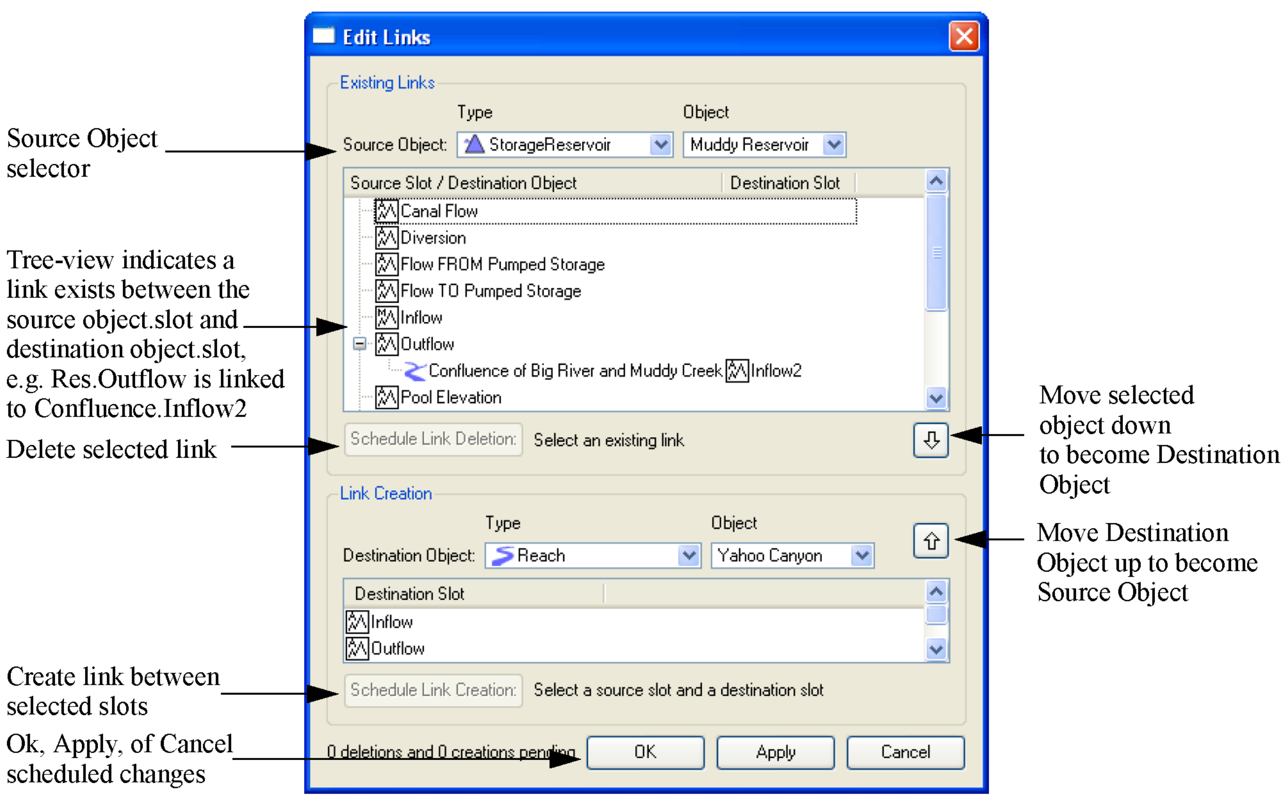
You can use either of the following methods to open the Edit Links tool.

- On the workspace menu, select **Workspace**, then **Edit Links**.
- On the workspace toolbar, select **Link** , then **Edit Links**.

About the Edit Links Tool

Figure 1.4 shows the Edit Links tool, which is divided into two areas: one for Existing Links and one for Link Creation.

Figure 1.4 Main parts of the Edit Links tool



Viewing and Deleting Existing Links

The Existing Links area of the Edit Link tool is used to view and delete existing links.

To generate a list of slots, on the **Source Object** menus, select an object **Type** and an **Object**. All linkable slots on that object are then listed in the Source Slot tree view. For aggregate objects, the upper-level aggregate and each element are listed; the expander arrow (>) next to each element allows you to expand the element to show the member slots, similar to the Object Viewer (or Open Object) tool.

The expander arrow (>) next to a slot name indicates the slot has a link. You can select the expander arrow (>) to expand the tree view and show the Destination Object and Destination Slot for all slots linked to the given slot.

If you highlight a row that includes a Destination Object and Destination Slot, the **Schedule Link Deletion** button becomes active. Selecting this button *schedules* the link for deletion; the link is not actually deleted until you confirm the deletion by selecting **Ok** or **Apply**. When a link is scheduled for deletion, the link name is displayed in strikethrough font, and there is an explanatory note next to the Schedule Link Deletion button.

> **Note:** You can create a link between two slots and then change methods on one or both objects so the linked slots are no longer visible or in use. The link remains on the workspace, and in the Edit Links tool, the Source Slot is shown in *italics* to indicate that it is not currently visible in the model. You cannot create links between slots that are not visible, but you can view and delete links between them.

Creating Links

Both the Existing Links and Link Creation areas of the Edit Link tool are used to create new links. First, on the **Source Object** menus, select an object **Type** and an **Object**. Then on the **Destination Object** menus, select an object **Type** and an **Object**. Select the source and destination slots in the upper and lower lists, then select **Schedule Link Creation** to *schedule* the link for creation. The link is created when you confirm by selecting **OK** or **Apply**. A link scheduled for creation is shown in green text, and the tree view is automatically expanded.

> **Note:** Links can be created between any pair of slots when one slot is listed in the Existing Links area of the tool and the other slot is listed in the Link Creation area. Some links may not make physical sense, but any combination can be defined.

Moving Objects Between Source and Destination

You may want to use the Edit Link tool to create a series of links in a specified order, such as upstream to downstream. To accomplish this, you can use the **Up** arrow  to move a Destination Object in the Link Creation area up, to become the Source Object.

Similarly, you can use the **Down** arrow  to move a Source Object down, to become the Destination Object. When you use this button, the object in the selected row is moved down. Thus, you can either select a slot (or make no selection) and the Source Object moves down; or you can highlight the linked Destination Object and Destination Slot row (through the tree view) and select the **Down** arrow to move the selected Destination Object down, to become the object selected in the Link Creation area.